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# Equivalence of volume, enthalpy and viscosity relaxation kinetics in glass-forming silicate liquids

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## Abstract

Structural relaxation in the glass transition range has been investigated for a series of silicates, aluminosilicates, borosilicates and titanosilicates. In complement to viscometry and drop calorimetry, a newly built instrument has been used for making simultaneous differential thermal analysis and dilatometry measurements. For all materials, the various glass transition temperatures correlate linearly with each other, indicating that the kinetics are the same for viscosity, enthalpy and volume relaxation in the glass transition range. This conclusion has been confirmed by measurements of the actual relaxation kinetics for viscosity and volume made for the so-called E borosilicate glass. That the same volume is observed starting from fictive temperatures higher and lower than the run temperature finally demonstrate that equilibrium volume measurements can be made over significant temperature intervals in the glass transition range, allowing the thermal expansion coefficient of supercooled liquids to be determined accurately. © 2002 Elsevier Science B.V. All rights reserved.

## 1. Introduction

The physical properties of amorphous substances depend measurably on thermal history in the glass transition range. Although interest has long been paid to structural relaxation (e.g. [1]), relatively few experimental studies have been devoted to determining the time dependence of physical properties in the glass transition range of the industrially and geologically important silicates. Particularly problematic is the fact that

different samples have generally been investigated by different techniques, so that the basic question of possible differences in relaxation kinetics between different physical properties has not been properly settled. Moynihan et al. [2] stressed that the mechanisms controlling enthalpy and volume relaxation could differ, but it is generally assumed that the relaxation kinetics are similar for different properties. For instance, this simplifying assumption has been used to determine thermal expansion coefficients of supercooled liquids by a combination a dilatometry and calorimetry experiments [3]. While the present work was in progress, support for this procedure has later been brought by Webb and Knoche [4] with a compilation showing a close correspondence between

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glass transition temperatures of silicates as determined from differential scanning calorimetry, dilatometry and viscometry.

In this study, our first aim has been to compare in a consistent manner the volume, enthalpy and viscosity relaxation kinetics. A wide range of simple and more complex silicates with varying degrees of polymerization has been considered, including liquids having anomalous thermal and relaxation properties like borosilicates [5,6] and titanosilicates [7,8], which make them especially interesting in this respect. The volume and enthalpy glass transition temperatures ( $T_g$ ) of all these materials have been simultaneously measured with a newly built apparatus and compared with the temperature at which the viscosity is  $10^{12}$  Pa s, which is the operational definition of glass transition for our standard heating rate of a few K/min.

Although important, glass transition temperatures as determined for a given heating rate give an incomplete description of relaxation. Unless thermorheological simplicity is assumed, that similar glass transition temperatures are observed for different properties for the same heating rate does not imply that relaxation kinetics are the same at any temperature. To complement this study, we have thus compared our results to the glass transition temperatures determined for much higher cooling rates from drop calorimetry measurements made previously in our laboratory on the same materials. As a direct investigation of relaxation kinetics, we have finally measured the time dependence of viscosity and volume in the case of E

glass, an industrially important borosilicate material. A noteworthy implication of these experiments is the demonstration that the equilibrium volume and thermal expansion coefficient can be measured accurately for highly viscous liquids.

Note finally that this paper is the last installment of a series relying on an extensive set of high-viscosity measurements. In two other contributions, we have studied [9] the connection between the viscosity–temperature relationships given either theoretically by the Adam–Gibbs [10] theory or empirically by Tammann–Vogel–Fulcher equations, and investigated viscosity relaxation times after sudden temperature or stress changes [11].

## 2. Experimental methods

All the materials investigated are synthetic. Their nominal compositions are listed in Table 1. The titanosilicates and E, NC and BNC glasses are from the same batches as the products studied by calorimetry and viscometry [5,6,8]. The other starting materials were prepared in a similar way from reagent grade oxides and carbonates dried at 1100 °C ( $\text{SiO}_2$ ,  $\text{MgO}$ ,  $\text{TiO}_2$ ), 550 °C ( $\text{CaO}$ ) or 350 °C ( $\text{Na}_2\text{O}$ ,  $\text{K}_2\text{O}$ ). The mixes were ground under alcohol in an agate mortar, slowly decarbonated in Pt crucibles in an electric muffle furnace and melted until bubble-free materials were obtained. Fluid melts were stirred to ensure chemical homogeneity. For viscous liquids like  $\text{NaAlSi}_3\text{O}_8$ , the

Table 1  
Chemical composition and room temperature density of the materials investigated (mol%)

|                  | $\text{SiO}_2$ | $\text{CaO}$ | $\text{MgO}$ | $\text{Al}_2\text{O}_3$ | $\text{Na}_2\text{O}$ | $\text{K}_2\text{O}$ | $\text{B}_2\text{O}_3$ | $\text{TiO}_2$ | $\text{FeO}$ | $\rho$ (g/cm <sup>3</sup> ) |
|------------------|----------------|--------------|--------------|-------------------------|-----------------------|----------------------|------------------------|----------------|--------------|-----------------------------|
| NS <sub>4</sub>  | 80.0           |              |              |                         | 20.0                  |                      |                        |                |              | 2.382                       |
| Ca 0.39          | 0.0            | 39.0         |              | 61.0                    |                       |                      |                        |                |              | 2.905                       |
| Ca 42.14         | 42.86          | 42.86        |              | 14.3                    |                       |                      |                        |                |              | 2.836                       |
| Ca 50.25         | 50.0           | 25.0         |              | 25.0                    |                       |                      |                        |                |              | 2.690                       |
| Ca 12.44         | 11.80          | 44.10        |              | 44.10                   |                       |                      |                        |                |              | 2.860                       |
| Di               | 50.0           | 25.0         | 25.0         |                         |                       |                      |                        |                |              | 2.852                       |
| Ab               | 75.0           |              |              | 12.5                    | 12.5                  |                      |                        |                |              | 2.368                       |
| NTS <sub>4</sub> | 66.67          |              |              |                         | 16.67                 |                      |                        | 16.67          |              | 2.634                       |
| KTS <sub>2</sub> | 50.0           |              |              |                         |                       | 25.0                 |                        | 25.0           |              | 2.666                       |
| NC               | 70.92          | 9.07         | 5.30         | 0.35                    | 13.56                 | 0.13                 |                        |                | 0.64         | 2.498                       |
| BNC              | 64.98          | 7.83         | 4.63         | 1.91                    | 15.45                 | 1.04                 | 3.94                   | 0.05           | 0.17         | 2.528                       |
| E                | 59.30          | 20.64        | 4.15         | 8.84                    | 0.46                  | 0.23                 | 5.81                   | 0.09           | 0.48         | 2.584                       |

material was quenched, reground and remelted several times. Densities measured at room temperature with an Archimedeian method (Table 1), toluene being used as the immersion fluid, were the same for different chips of the same glass. Likewise, the chemical analyses measured with an electron microprobe did not show either significant variations for a given glass or deviations from the nominal compositions. More detail on this and other experimental aspects are given by Sipp [12].

The volume and enthalpy glass transition temperatures have been measured simultaneously with a differential thermal analysis (DTA)-dilatometer apparatus built in our laboratory [12]. Differential measurements are made between the sample and a silica cylinder, both being about 10 mm in height and 5 mm in diameter, enclosed in the silver chamber depicted in Fig. 1. This chamber is placed in the hot spot of the furnace made of two fibro-

thal half shells whose temperature is set and programmed with a digital controller. Isothermal conditions are ensured by the high thermal conductivity of silver and proper adjustment of the chamber position within the furnace. Under 'isothermal' conditions, the temperature gradient within the silver chamber is smaller than 1 K.

The lengths of sample and reference are measured as a function of time to within 0.1  $\mu\text{m}$  from the displacement of silica rods with two linear voltage differential transducers (LVDTs) whose springs have been removed to reduce the force exerted on the sample and thus limit micropenetration of the measuring silica rod in the sample. For accuracy and precision, the temperature within the chamber is measured with a Pt/PtRh10% thermocouple made of 0.5 mm wires. To maximize DTA sensitivity, the temperature difference between the sample and silica reference is measured with chromel–alumel thermocouples, passing through the supporting alumina plates, whose hot junctions are inserted into small conical holes drilled at the base of the sample and reference. Thanks to the sensitivity of the apparatus, all kinetic measurements could be made at the relatively low rate of 2.2 K/min. This allowed temperature gradients in the rather large sample and silica reference to be minimized during the heating stage of the experiments. For isothermal length measurements, the temperature was stabilized at the run temperature in less than 15 min from the previous  $T$ .

For all experiments, the temperature and lengths are recorded with a computer through an IEEE interface. As usual, the dilatometric glass transition temperature ( $T_{\text{Dil}}$ ) has been defined as the temperature at which the linearly extrapolated lengths of the glass and supercooled liquids intersect (Fig. 2(a)). Depending on the way they are defined in a given thermogram, DTA glass transition temperatures ( $T_{\text{DTA}}$ ) can differ by 50 K, which corresponds to variations of one and a half order of magnitude in viscosity. Here, the DTA glass transition temperature was defined as the inflexion point on the thermogram (Fig. 2(b)). Both  $T_{\text{Dil}}$  and  $T_{\text{DTA}}$  are precise to better than 3 K.

Viscosities were measured under compressional stress with the apparatus described by Neuville

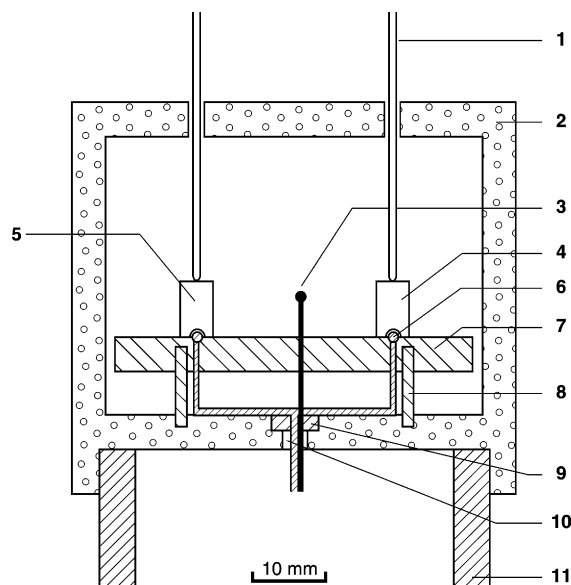


Fig. 1. High-temperature chamber of the dilatometry-DTA apparatus. 1: silica rods (to LVDTs); 2: silver chamber; 3: Pt/Pt-Rh10% measuring thermocouple; 4: sample; 5: silica glass reference; 6: hot junction of the differential chromel/alumel thermocouple; 7: alumina plate supporting sample and reference; 8: alumina cylinder supporting the alumina plate; 9: alumina separator ensuring electrical insulation of thermocouple wires; 10: four-hole alumina sheath for thermocouples; 11: alumina tube supporting the chamber in the furnace.

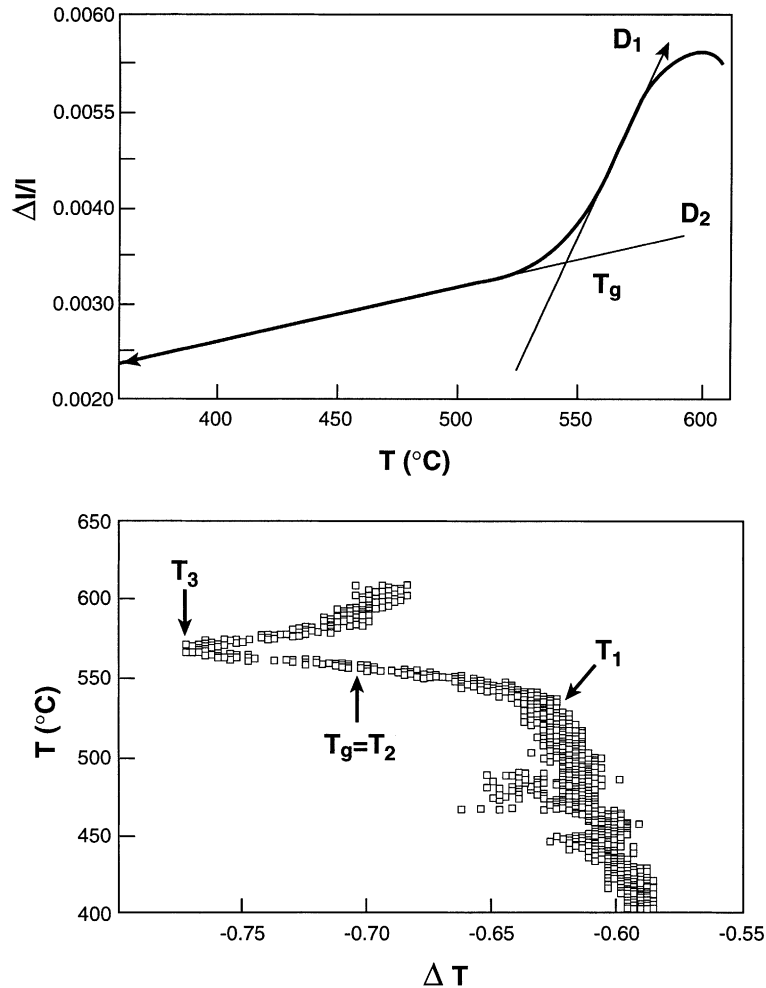


Fig. 2. Glass transition ( $T_g$ ) of BNC (a) as determined from dilatometry and (b) differential thermal analysis.

and Richet [13] as adapted by Sipp et al. [6] for extending the measurements down to the range  $10^{12}$ – $10^{15}$  Pa s. With the cylindrical geometry of our samples, the viscosity ( $\eta$ ) is obtained from the strain rate ( $\dot{\epsilon}$ ) measured at constant stress ( $\sigma$ ) by

$$\eta = \sigma / 3\dot{\epsilon}. \quad (1)$$

Viscosities at low temperatures were first measured up to about  $10^{12}$  Pa s, i.e., under conditions where relaxation times were short enough that equilibrium was achieved during the 15–30 min required to ensure a constant sample temperature after the set point had been changed. The average of 10–15 measurements made at different stresses,

spanning in general at least one order of magnitude at a given temperature, have typical standard errors smaller than 0.02 log units which are similar to the differences found with the data reported for NBS standard glasses. In all cases viscosities were thus found to be Newtonian. Because of the long time required to reach equilibrium at lower temperatures, measurements were then usually made for a single stress of about 200 MPa down to about  $10^{14}$  Pa s. As described by Sipp [12], the same results were found for a few materials for which different series of experiments were made at different stresses. For the temperature  $T_{12}$  at which the viscosity is  $10^{12}$  Pa s, the error is less than 2 K.

Comparisons between glass transition temperatures will also be made with drop calorimetry data. These have the interesting feature of referring to glass transitions that took place at faster cooling rates of about 100 K/min around 1100 K [14]. As usual, the drop calorimetry glass transition temperatures ( $T_{\text{DC}}$ ) are given by the intersection of the mean heat capacity curves of the glass and liquid. The uncertainties of about 10 K on  $T_{\text{DC}}$  are thus much greater than for the other techniques.

### 3. Results

#### 3.1. Viscosity

The newly measured viscosities are listed in Table 2 for Ca 0.39, Ca 42.14 and NS<sub>4</sub>. They are plotted in Fig. 3 where they are compared with previous measurements for the same materials [13,15,16]. For NC and BNC, titanosilicates and the other liquids the viscosities have been reported by Sipp et al. [6,9] and Bouhifd et al. [8], respectively. Contrary to the commonly held view that viscosity reverts to Arrhenian variations in the glass transformation range, deviations from Arrhenian variations of the viscosity are observed in all cases. Up to the highest equilibrium values measured in study, namely  $10^{15}$  Pa s, these are accounted for with Tammann–Vogel–Fulcher equations:

$$\log \eta = A + B/(T - T_1), \quad (2)$$

where  $A$ ,  $B$  and  $T_1$  are adjustable parameters listed in Table 3. Good results are also obtained in terms of the configurational entropy theory of Adam and Gibbs [10] when including low viscosities at superliquidus temperatures into the fits. Since in this work we are mainly interested in glass transition temperatures, we refer to Sipp et al. [9] for a description of these other fits and for a discussion of the significance of the high-viscosity data.

#### 3.2. Glass transition temperatures

The observed DTA and dilatometry glass transition temperatures are listed in Table 4. Also

Table 2  
Measured viscosities (Pa s)

| $T$ (K)               | $\log \eta$ |
|-----------------------|-------------|
| <i>NS<sub>4</sub></i> |             |
| 703.3                 | 14.50       |
| 717.9                 | 13.75       |
| 726.4                 | 13.32       |
| 727.0                 | 13.29       |
| 740.1                 | 12.58       |
| 753.5                 | 11.98       |
| 760.8                 | 11.63       |
| 774.3                 | 11.05       |
| 787.7                 | 10.58       |
| 801.4                 | 10.08       |
| 815.2                 | 9.56        |
| 815.8                 | 9.54        |
| 823.7                 | 9.33        |
| 828.1                 | 9.20        |
| 836.0                 | 8.98        |
| <i>Ca 0.39</i>        |             |
| 1085.7                | 14.41       |
| 1092.2                | 13.98       |
| 1109.1                | 12.67       |
| 1109.9                | 12.65       |
| 1119.6                | 11.96       |
| 1128.9                | 11.35       |
| 1148.5                | 10.23       |
| 1154.6                | 9.92        |
| 1165.7                | 9.45        |
| <i>Ca 42.14</i>       |             |
| 1029.2                | 14.20       |
| 1029.4                | 14.12       |
| 1038.2                | 13.55       |
| 1046.1                | 13.07       |
| 1047.2                | 13.03       |
| 1049.2                | 12.82       |
| 1064.7                | 12.06       |
| 1072.0                | 11.61       |
| 1084.7                | 11.08       |
| 1084.9                | 10.92       |
| 1100.7                | 10.31       |
| 1100.8                | 10.17       |
| 1113.2                | 9.78        |
| 1113.5                | 9.65        |
| 1120.6                | 9.42        |

included are the viscometry temperatures  $T_{12}$ , calculated with the data of Table 3, and the drop-calorimetry temperatures  $T_{\text{DC}}$  determined in our laboratory [5,13,17,18]. Comparisons between dilatometry and DTA glass transition temperatures show an almost perfect linear correlation (Fig. 4). Interestingly, the slope of the relationship

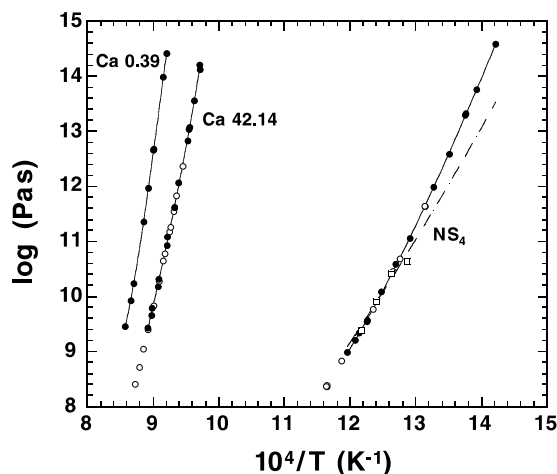


Fig. 3. Viscosity of Ca 0.39, Ca 12.44 and NS<sub>4</sub>. For comparison, the data of Neuville and Richet ([13], open circles) for Ca 42.14 and Poole ([15], open circles) and Knoche et al. ([16], open squares) for NS<sub>4</sub> have been included. The extrapolation of the fit reported for NS<sub>4</sub> by Knoche et al. (1995), shown as a dashed line, results in a 12 K error on  $T_{12}$  or of 0.51 log units on  $\eta(T_g)$ .

is slightly greater than 1, the difference between  $T_{DTA}$  and  $T_{Dil}$  increasing slightly with decreasing temperatures. Both temperatures are the same for high  $T_g$ s, whereas  $T_{DTA}$  is about 10 K greater than  $T_{Dil}$  for  $T_g$ s of about 800 K. This effect originates in the fact that these temperatures are not defined from the same criterion (Fig. 2). Albeit small, it

Table 3

Coefficients of  $\log \eta = A + B/(T - T_1)^a$

|                               | $A$   | $10^{-3} B$ | $T_1$              |
|-------------------------------|-------|-------------|--------------------|
| NS <sub>4</sub>               | −2.65 | 4702        | 431.3              |
| Ca 0.39                       | −4.51 | 4156        | 867.3              |
| Ca 42.14                      | −3.13 | 4099        | 793.2 <sup>b</sup> |
| Ca 76.11 <sup>c</sup>         | −4.75 | 8466        | 648.7              |
| Ca 50.25 <sup>c</sup>         | −4.74 | 5811        | 781.9              |
| Ca 12.44 <sup>c</sup>         | −4.56 | 4230        | 881.7              |
| Di <sup>c</sup>               | −4.19 | 4362        | 724.6              |
| Ab <sup>c</sup>               | −5.58 | 13011       | 345.8              |
| NTS <sub>4</sub> <sup>d</sup> | −2.82 | 3415        | 640.9              |
| KTS <sub>2</sub> <sup>d</sup> | −3.00 | 2915        | 671.3              |
| NC <sup>e</sup>               | −3.01 | 4747        | 504.3              |
| BNC <sup>e</sup>              | −3.00 | 4207        | 538.9              |
| E <sup>e</sup>                | −4.82 | 6606        | 551.9              |

<sup>a</sup> Temperatures in K, viscosities in Pa s.

<sup>b</sup> Fit made with the data of Neuville and Richet [13].

<sup>c</sup> From the data of Sipp et al. [9].

<sup>d</sup> Bouhifd et al. [8].

<sup>e</sup> Sipp et al. [6].

underlines the importance of using consistent observations for comparing relaxation kinetics between different properties.

Although  $T_{DTA}$  and  $T_{12}$  were not simultaneously measured, their mutual consistency is also excellent (Fig. 5). The reason is of course that viscosity is an equilibrium property so that possible differences in thermal history between the same samples investigated in the two kinds of experiments

Table 4

Glass transition temperatures<sup>a</sup> (K) and viscosity at these temperatures (log Pa s)

|                  | $T_{Dil}$ | $T_{DTA}$ | $T_{DC}$ | $T_{12}$ | $\eta(T_{Dil})$ | $\eta(T_{DTA})$ | $\eta(T_{DC})$ |
|------------------|-----------|-----------|----------|----------|-----------------|-----------------|----------------|
| NS <sub>4</sub>  | 751       | 758       |          | 753      | 12.07           | 11.76           |                |
| Ca 0.39          | 1123      | 1125      |          | 1119     |                 | 11.62           |                |
| Ca 42.14         | 1072      | 1073      | 1085     | 1065     | 11.64           | 11.57           | 10.98          |
| Ca 76.11         | 1163      | 1164      | 1199     | 1155     |                 | 11.70           | 10.66          |
| Ca 50.25         | 1141      | 1138      | 1160     | 1130     |                 | 11.62           | 10.67          |
| Ca 12.44         | 1146      | 1147      |          | 1137     |                 | 11.38           |                |
| Di               | 999       | 1005      | 1005     | 994      | 11.72           | 11.38           | 11.38          |
| Ab               | 1095      | 1097      | 1096     | 1085     | 11.76           | 11.71           | 11.74          |
| NTS <sub>4</sub> | 881       | 885       | 904      | 873      | 11.48           | 11.24           | 10.22          |
| KTS <sub>2</sub> | 865       | 873       |          | 865      | 12.03           | 11.42           |                |
| NC               | 818       | 828       | 850      | 821      | 12.14           | 11.67           | 10.74          |
| BNC              | 817       | 829       | 850      | 820      | 12.15           | 11.54           | 10.58          |
| E                | 952       | 955       | 996      | 945      | 11.75           | 11.60           | 10.10          |
| Mean             |           |           |          |          | 11.86           | 11.58           | 10.78          |
| Std dev          |           |           |          |          | 0.24            | 0.11            | 0.52           |

<sup>a</sup> From dilatometry, differential thermal analysis, drop calorimetry and viscometry, respectively.

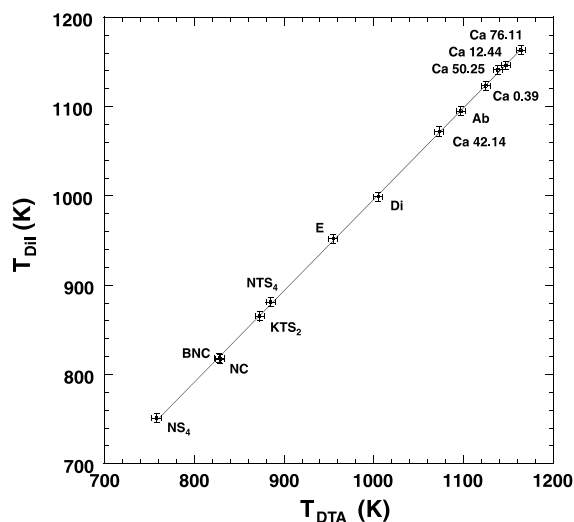


Fig. 4. Dilatometry against DTA glass transition temperatures.

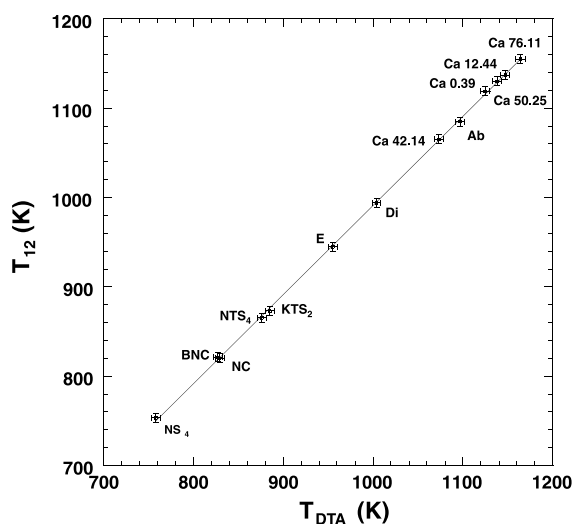


Fig. 5. Viscometry against DTA glass transition temperatures.

cannot induce any scatter in this comparison. This is not the case for the drop calorimetry glass transition temperature. The consistency of  $T_{DC}$  with  $T_{DTA}$  (Fig. 6),  $T_{Dil}$  and  $T_{12}$  is thus less good because of the higher uncertainties besetting  $T_{DC}$  and especially the fact that different samples with different thermal histories have been investigated. Consistent with the differences between the rates of

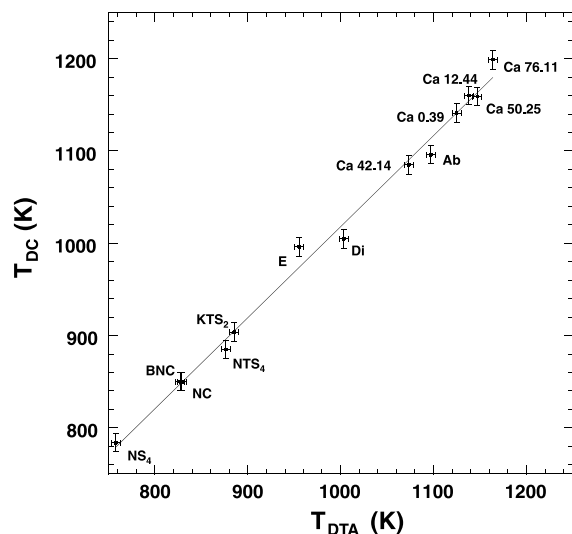


Fig. 6. Drop calorimetry against DTA glass transition temperatures.

both techniques,  $T_{DC}$  is systematically higher than  $T_{DTA}$  by about 20 K.

### 3.3. Relaxation

We finally plot in Fig. 7 the viscosity and volume relaxation observed for E glass under similar temperature–time conditions. This material was chosen in this first investigation because of the fast

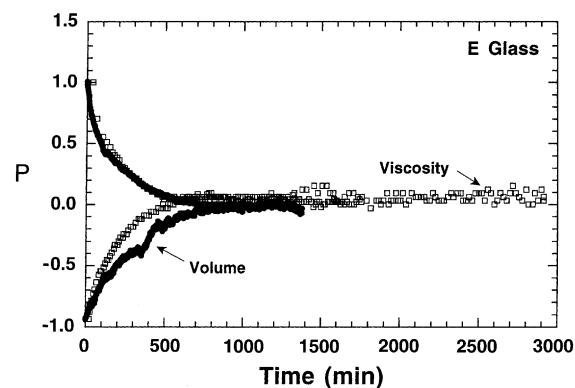


Fig. 7. Comparison between volume and viscosity relaxation for E glass at 910 K, after equilibration at 967 K (upper curves) and 900 K (lower curves), in the form of normalized parameters  $P = (P_t - P_\infty)/(P_0 - P_\infty)$ . Viscosity measurements reported by Sipp et al. [6] and new volume data.

relaxation previously observed in viscosity measurements at the lower end of the glass transition range [6]. After equilibration at 967 K (i.e., for a viscosity of  $10^{11.1}$  Pa s), the sample temperature was rapidly lowered to 911 K. The length decreased by 6  $\mu\text{m}$  until stability was achieved in about 700 min. The temperature was then decreased to 900 K, at which a constant length was observed after about 1000 min. The temperature was finally increased back to 911 K where a constant length was again reached in about 700 min after a 2  $\mu\text{m}$  increase. As the same length was measured starting from fictive temperatures either higher or lower than the run temperature, the measurements demonstrate that equilibrium was actually reached.

To compare in Fig. 7 the observed relaxation kinetics for volume and viscosity we have followed the common practice of plotting the experimental results as

$$P = (P_t - P_\infty)/(P_0 - P_\infty), \quad (3)$$

where  $P_t$ ,  $P_\infty$  and  $P_0$  are the values of property  $P$  at time  $t$ , at equilibrium, and at time 0, respectively. Of these three values  $P_0$  is the least known because relaxation was already taking place during the time required to reach thermal equilibrium in the furnace before the measurements were started. This was more of a problem in runs where the fictive temperature was increasing because of faster initial kinetics, and for viscometry than for dilatometry because of longer times required to reach thermal equilibrium in the furnace. Given this uncertainty, the consistency of both measurements is striking. Hence the other important feature is that the kinetics are the same for volume and viscosity even when the relaxation is very slow.

## 4. Discussion

### 4.1. Shear moduli and Maxwell model

The viscosities at the various glass transition temperatures are included in Table 4. At  $T_{\text{DTA}}$ , for instance, we find that the average viscosity is  $10^{11.58}$  Pa s. This mean value has no particular signifi-

cance in view of the arbitrary choice of the inflexion point in thermograms for defining  $T_{\text{DTA}}$ . In contrast it is worth noting that there is only half an order of magnitude difference between the extreme values of  $10^{11.24}$  and  $10^{11.76}$  Pa s and that the standard deviation from the mean is only  $10^{0.11}$  Pa s. These figures contrast with the wider interval  $10^{9.71}$ – $10^{12.08}$  Pa s and greater standard deviation of  $10^{0.33}$  Pa s reported by Webb and Knoche [4] for the viscosity at the glass transition of 70 silicates related to granite compositions or belonging to the systems  $\text{Na}_2\text{O}$ – $\text{SiO}_2$  and  $\text{NaAlSi}_3\text{O}_8$ – $\text{CaAl}_2\text{Si}_2\text{O}_8$ – $\text{CaMgSi}_2\text{O}_6$ . As indicated by higher scatter in  $T_g$ – $T_g$  plots, the greater spread of the values reported by Webb and Knoche [4] results from a less good accuracy and the resulting uncertainties due to the extrapolation of measurements made to calculate the viscosity at the glass transition temperature (see Fig. 3). Likewise, the  $10^{10.2}$ – $10^{11.2}$  Pa s range first noted by Neuville and Richet [13] for the drop calorimetry glass transition temperature or the new  $10^{10.1}$ – $10^{11.7}$  Pa s interval indicated by the data of Table 4, are too large because of the aforementioned uncertainties on  $T_{\text{DC}}$ .

The width of the viscosity interval at the glass transition is an important feature because it gives a measure of the variation with composition of the shear modulus at infinite frequency  $G_\infty$  through the Maxwell relation

$$\eta(T) = \tau(T)/G_\infty(T), \quad (4)$$

where  $\tau$  is the relaxation time. For silicates, a review of available data has suggested that  $G_\infty$  weakly depends on composition, with values in the range 3–32 GPa [19]. Since  $\tau$  is the same for different materials heated or cooled at the same rate, the data of Webb and Knoche [4] for dilatometric  $T_g$  values in the range 710–1160 K yield a spread of 2.4 orders of magnitude for  $G_\infty$  at the glass transition. In contrast, the results of Table 4 indicate a much narrower range of only half an order of magnitude despite the fact that  $T_g$  varies from 750 to 1160 K. The weak composition dependence of  $G_\infty$  emphasized by Dingwell and Webb [19] is thus confirmed by the present results.

All together our results do not show any clear trend between  $T_g$  and  $\eta(T_g)$ , even for related compositions. This is apparent in Fig. 8 by the lack

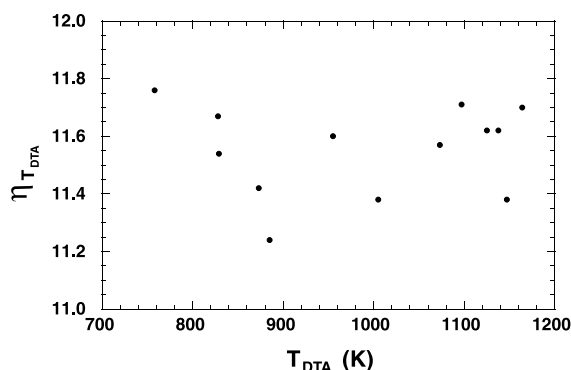


Fig. 8. Viscosity at the DTA glass transition temperature against  $T_{DTA}$ .

of correlation between  $T_{DTA}$  and  $\eta(T_{DTA})$ . Hence there is currently no simple way of separating from such results the composition and temperature dependences of  $G_{\infty}$ , if any. Since  $G_{\infty} = \rho v_s^2$ , where  $\rho$  is the density and  $v_s$  the velocity of shear acoustic waves at infinite frequency, the temperature dependence of  $G_{\infty}$  can be directly determined from Brillouin scattering observations because these refer to extremely short time scales of about  $10^{-10}$  s. From Brillouin data above the glass transition and density data for pure  $\text{SiO}_2$  [20], Di and Ca 50.25 melts [21,22], we actually calculate that  $G_{\infty}$  is in the range 34–36 GPa at  $T_g$  and that  $dG_{\infty}/dT$  is only of the order of  $-0.02$  GPa/K. In the 400 K interval spanned by the data of Fig. 8,  $G_{\infty}$  should decrease by less than 8 GPa and remain close to the 3–32 range assumed by Dingwell and Webb [19].

We thus emphasize that, although we have considered fewer compositions than Webb and Knoche [4], our results deal with a wider composition interval so that they actually provide a tighter upper bound to differences in viscosity and  $G_{\infty}$  at the glass transition. This is especially so as our database includes a silica-free melt (Ca 0.39) as well as titanosilicates and borosilicates which show peculiar viscosity and entropy effects due to unusual increases of the configurational heat capacity when the glass transition is approached [5,7,8].

These relatively small differences in shear moduli at infinite frequency nevertheless translate in significant differences in relaxation kinetics. For

instance, the measurements reported by Sipp et al. [6] and Bouhifd et al. [8] show that the time required for equilibrium to be reached can vary by a factor of 3 for materials having the same viscosity and thermal histories. The Adam–Gibbs relaxation theory predicts that the relaxation time is inversely proportional to the configurational entropy of the melt ( $S^{\text{conf}}$ ). From calorimetrically determined values of  $S^{\text{conf}}$ , one can account for the temperature dependence of the equilibrium viscosity of silicate melts with only two fit parameters [23]. Far from equilibrium, experience interestingly shows that the relaxation kinetics are the slowest for materials which, like titanosilicates [5] and some borosilicates [8], lose unusually rapidly configurational entropy. The manner in which the irreversible aspect of relaxation can be simply predicted from configurational entropy remains, however, to be established.

#### 4.2. Viscosity vs. volume and enthalpy relaxation

The equivalence of the kinetics of enthalpy and volume relaxation in glass-forming liquids has long been debated. For poly (vinyl acetate), for example, the enthalpy and volume relaxation kinetics are characterized by different relaxation times [24] and the viscosities of the polymers investigated by Plazek et al. [25] differ at the dilatometric glass transition temperatures. Although slight differences in these kinetics have been reported for  $\text{As}_2\text{Se}_3$  [2], the viscosity and enthalpy relaxation appear equivalent for  $\text{B}_2\text{O}_3$  [26]. Likewise, Sasabe et al. [27] showed for the NBS 710 standard borosilicate glass the equivalence of the relaxation kinetics for enthalpy, viscosity and refractive index from independently made DSC, viscosity and refractive index measurements. These results are thus consistent with the direct observations made by Rekhson et al. [28] that the viscosity and volume of another silicate, window glass, relax at the same rate.

Extending the composition range investigated, a similar equivalence of relaxation kinetics was observed by Webb and Knoche [4] from a comparison of the glass transition temperatures they determined from viscometry, calorimetry and DSC observations. As noted in the previous section, the

simultaneous nature of the dilatometric and DTA measurements made in the present work eliminates any bias that could beset such comparisons. The correlations of Figs. 4–6 are thus all the more significant. In addition, the measurements on E glass directly demonstrate the equivalence of the kinetics of volume and viscosity relaxation since, in both cases, equilibrium is reached after the same length of time (Fig. 6). For this borosilicate, the conclusion drawn by Rekhson et al. [28] in the case of window glass is thus confirmed for longer time scales. From a practical standpoint, an important consequence is that volume relaxation can be modeled from volume measurements or vice versa, which also limits the number of parameters needed to model relaxation. As a matter of fact, Scherer [29] successfully accounted for in this way enthalpy and refractive index measurements for NBS 710 glass by fitting the model parameters to available viscosity data.

In view of the similarity of structural relaxation time and Si–O average lifetime as determined by nuclear magnetic resonance spectroscopy [30], it is now generally agreed that rearrangement of Si–O bonds is the critical step governing both relaxation and viscous flow of silicates. For instance, Webb and Knoche [4] assigned to this mechanisms the similarity of various relaxation kinetics. Since the correlation time of the quadrupolar relaxation mechanism for aluminum also scales with the viscosity relaxation time [31,32], this mechanism must also operate in aluminates, like Ca 0.39, whereas the same microscopic fluctuations must involve both Si–O and Al–O bond exchange in aluminosilicates like Ca 42.14.

#### 4.3. Thermal expansion of glasses and supercooled liquids

Even though the temperature intervals over which length measurements can be made are not wide, another implication of the present results is the possibility of measuring accurately the equilibrium volume of supercooled liquids in the glass transition range. These intervals are limited to viscosities of about  $10^{14}$  Pa s at low temperatures by the slowness of relaxation kinetics and, at higher temperatures, by incipient softening of the

sample. In practice, this corresponds to temperature intervals of about 50 K. Thanks to the good precision of the measurement of the length ( $l$ ) of samples, the thermal expansion coefficients of supercooled liquids ( $\alpha_l$ ) can nonetheless be determined with an error of less than 5% from  $\alpha_l = 3(\partial \ln l / \partial T)_p$ . By measuring the room-temperature density of the glass and the length of the glass up to the glass transition temperature, one can also determine the density ( $\rho$ ) and volume of the supercooled liquid. A practical problem stems from micropenetration in the sample of the rod used for length measurements. As described by Toplis and Richet [22], this problem can be taken care of simply as long as the sample viscosity is higher than about  $10^{12}$  Pa s.

Measurements of thermal expansion coefficients in the glass transition range are in fact more straightforward for the supercooled liquid than for the glass. Because volume relaxation actually begins well below the dilatometric glass transition temperature (see [22]), the length measured below  $T_{Dil}$  does not refer to a constant configuration. The magnitude of this effect of course depends on the heating rate of the experiment. Unless the sample has been previously stabilized at the lower end of the glass transition range, the fictive temperature of the sample decreases so that the measured expansion represents a lower bound to the expansion at constant configuration.

Another useful application of these experiments is the determination of the compaction of quenched glasses after annealing in the glass transition range, which was discovered in 1845 by Chevan-dier and Wertheim [33]. The compaction is defined as

$$C = (\rho_0 - \rho_1) / \rho_0, \quad (5)$$

where  $\rho_0$  and  $\rho_1$  are the initial and final densities of the sample. These can be readily determined from the length changes of a sample equilibrated at two different temperatures. Alternatively, the compaction can be measured directly from room-temperature measurements of the density of glasses subjected to some heat treatment. Knowing the thermal expansion coefficients of the glass and supercooled liquid phases, the change in fictive

temperature could then be calculated from the observed compaction with the assumption that the thermal expansion coefficient of the glass is insensitive to thermal history. Note finally that in our dilatometric experiments the instantaneous, vibrational contribution to the dilation of the supercooled liquid can be distinguished from the slow configurational contribution shown in Fig. 7 [34]. It is then possible to determine independently the ‘glass’ part of the thermal expansion coefficient, to determine how it depends on the configurational state and check its consistency with the value determined below  $T_g$  for the glass.

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